MIL-C-24643/5D
14 March 1994
SUPERSEDING
MIL-C-0024643/5C
29 May 1987
MIL-C-24643/5B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSMDU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated, (see table I for size) Coated copper may be used at manufacturer's option where required to provide free stripping insulation.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Ethylene propylene rubber or cross-linked polyethylene insulation or at the manufacturer's option a composite with a primary insulation consisting of ethylene propylene rubber or cross-linked polyethylene. The composite layer may be substituted for up to 10% of the wall thickness (see table I for wall thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.

* Fourth - Nineteen conductors, cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with no. 1 from center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

Fifth - Binder tape applied helically with overlap.

* Sixth - Cross linked polyolefin jacket. (see table I for wall thickness).

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* Table I. Details

Military part no. M24643/5	Type and size	Conductors --- Size AWG	Insulation thickness minimum average (inch)	Cable jacket thickness minimum average (inch)	Overall diameter maximum (inch)	Conductor resistance per 1000 feet maximum (ohms)	Insulation resistance per 1000 feet minimum (megohms)	Cold bending mandrel diameter max (inch)
-01UN	LSMDU-6	12(class B)	0.028	0.060	1.000	1.730	100	13
-02UN	LSMDU-14	9(class B)	.040	.060	1.395	.868	100	19
-03UN	LSMDU-23	7(class B)	.052	.075	1.765	.598	100	23
-04UN	LSMDU-40	4(class C)	.052	.075	2.040	.273	90	27
-05UN	LSMDU-60	2(class D)	.052	.075	2.330	.172	75	30

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum...	(see table I)
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2500
Insulation resistance - megohms/1000 feet, minimum.....	(see table I)
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage.....	No failure
* Group B:	
Cold bending, cable - (see table I for mandrel diameter).....	No damage
Drip - 95 ± 1°C.....	Zero
* Cross-linked proof test (percent, maximum)	
Insulation.....	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)...	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum.....	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160

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EXAMINATION AND TESTS (continued):

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
* Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	125
Permanence of printing (conductor) - cycles, minimum	25
Heat distortion - percent of unaged, maximum	30
Cable sealant removability	No failure
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2

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QUALIFICATION INSPECTION (continued):	Requirements
Immersion (jacket)	
Tensile strength - percent of unaged, minimum.....	50
Elongation - percent of unaged, minimum.....	50
Smoke index, maximum	
Jacket.....	25
* Fillers.....	45
Insulation.....	45
Toxicity index, maximum	
Jacket.....	5
Fillers.....	5
Insulation.....	1.5
Durometer (jacket) - (type A) hardness, minimum.....	80
Weathering (jacket).....	No failure

UNIT ORDERING LENGTH: 1000 feet nominal length for sizes 6, 14 and 23.
500 feet nominal for sizes 40 and 60.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project No. 6145-2057-05)

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